American Computer Science League

2025-2026 ● Contest 1: Short Problems ● Intermediate Division

1. Computer Number Systems

Evaluate the following and express the answer in octal.

$2025_8 - 202_8 + 20_8 / 2_8$

A. 1721
B. 1367
C. 1633
D. 1623
E. 1523

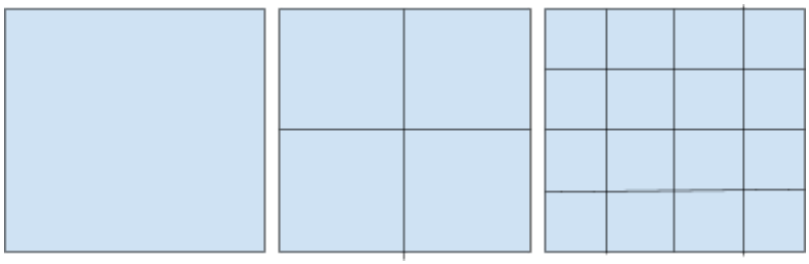
2. Computer Number Systems

The next number of the Fibonacci sequence is formed by adding the previous two numbers. The numbers are 0, 1, 1, 2, 3, 5, … How many times does 1 appear in the binary representation of the first 10 Fibonacci numbers starting at 0?

A. 4
B. 10
C. 16
D. 20
E. 22

3. Recursive Functions

Given a 16 cm by 16 cm square.
Repeatedly divide the square into 4 equally sized squares by connecting the midpoints of the sides vertically and horizontally.



Stop after subdividing the square 5 times.
What is the perimeter of the smallest square?

A. 0.5
B. 1
C. 2
D. 4
E. 8

4. Recursive Functions

Find $f(16, 12)$ given:

$$f(x,y) = \begin{cases} f(x-2, y+1) + y & \text{if } x > y \\ f(x+1, y) - x & \text{if } x < y \\ \dfrac{x}{2} & \text{if } x = y \end{cases}$$

A. 6
B. 7
C. 13
D. 14
E. 18

5. What Does This Program Do?

Which of the following input values of a, b, and c will produce an output of 7, 24, 25?

```
input a, b, c
if c == 0 then
     c = sqrt(a^2 + b^2)
else
    if b == 0 then
        b = sqrt(c^2 - a^2)
     else
        a = sqrt(c^2 - b^2)
     end if
end if
output a, b, c
```

A. 0, 24, 25
B. 7, 0, 25
C. 7, 24, 0
D. All of the above
E. None of the above

6. What Does This Program Do?

What is output by the following program?

```
a = 13
b = 21
n = 8
while sqrt(b) != int(sqrt(b))
     n = n + 1
     c = a + b
     a = b
     b = c
end while
output n
```

A. 12
B. 10
C. 11
D. 16
E. 13